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PHY - 101

TUTORIAL #1

Q1- a) $v_f^2 = v_i^2 + 2a(x_f - x_i)$

LHS $\rightarrow$ $(M^0L^1T^{-1})^2$

$= \underline{M^0L^2T^{-2}}$

RHS $\rightarrow$ $(M^0L^1T^{-1})^2 + 2(M^0L^1T^{-2})(L)$

$= \underline{M^0L^2T^{-2}}$

Hence, Proved.

b) If multiplied by 'M'

New dimension $\rightarrow$ $M^1L^2T^{-2}$

It is the dimension of ENERGY

Q2- a) $r = 6.5 \times 10^3$ km

Thickness = 10 km

$\therefore$ VOLUME = Surface area $\times$ Thickness

(Possible as thickness << S.A.)

$= 4\pi r^2 \times 10$

$= 40 \times 3.14 \times 6.5 \times 6.5 \times 10^6$

$= 40 \times 3.14 \times 42.25 \times 10^6$

$= 5.3 \times 10^9$ km^3

$= 5.3 \times 10^9 \times 10^9$ m^3

$= 5.3 \times 10^{18}$ m^3

Hence, best estimate $\rightarrow$ D) $\underline{(10^{15} - 10^{20})\,m^3}$

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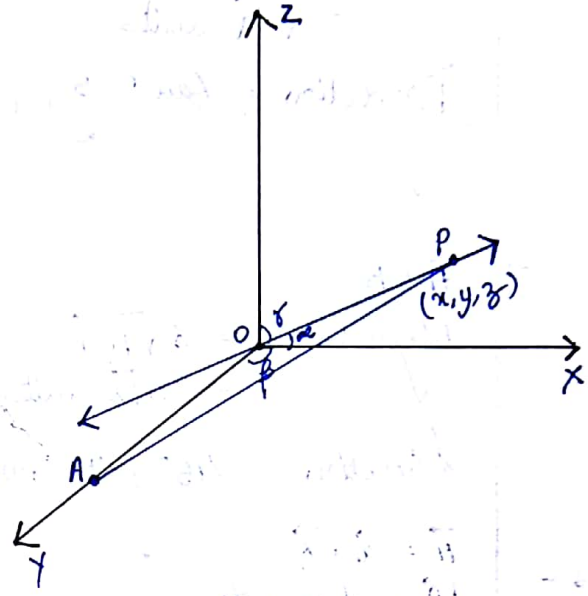
Tutorial #2

Q1 -



$$\cos\beta = \frac{OA}{OP} = \frac{y}{r}$$

$$\therefore y = r\cos\beta$$

Similarly,

$$x = r\cos\alpha$$

and

$$z = r\cos\gamma$$

$$x^2+y^2+z^2 = r^2(\cos^2\alpha + \cos^2\beta + \cos^2\gamma)$$

However,

$$x^2+y^2+z^2 = r^2$$

$$\Rightarrow \cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$$

Hence, Proved.

Q2 -

Using vector method

Diagonals:- $\vec{A}+\vec{B}$, $\vec{B}-\vec{A}$

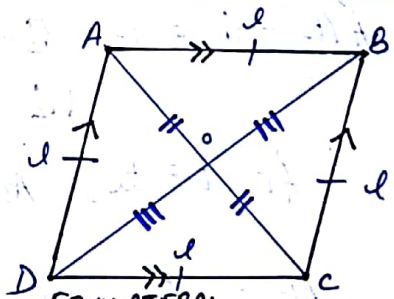
$(\vec{A}+\vec{B})\cdot(\vec{B}-\vec{A})$
$= \cancel{\vec{A}\cdot\vec{B}} + |\vec{B}|^2 - |\vec{A}|^2 - \cancel{\vec{B}\cdot\vec{A}}$
$= |\vec{B}|^2 - |\vec{A}|^2$

As $|\vec{B}| = |\vec{A}|$

$(\vec{A}+\vec{B})\cdot(\vec{B}-\vec{A}) = 0$

$\therefore$ DIAGONALS ARE PERPENDICULAR

To show: $AC \perp BD$



EQUILATERAL PARALLELOGRAM = RHOMBUS

We know: $AB = BC = CD = DA$
$AO = OC$
$BO = OD$
$AD \parallel BC$
$CD \parallel AB$

Sol^n: In $\triangle AOB$ and $\triangle COB$

S	$OB = OB$	($\because$ Common)
S	$AB = CB$	($\because$ Rhombus)
S	$AO = CO$	($\because$ Bisected)

$\therefore \triangle AOB \cong \triangle COB$ [$\because$ by SSS]

$\Rightarrow \angle AOB = \angle COB$ [$\because$ by CPCT]

As they are supplementary, $\angle AOB = \angle COB = 90^\circ$

Hence, Proved.

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Q4- a) $|\vec{A}| = \sqrt{25+4}$

$= \sqrt{29}$ units

Direction $= -\tan^{-1}\left(\frac{2}{5}\right)$ with +ve x-axis.

b) $|\vec{B}| = \sqrt{25+4}$

$= \sqrt{29}$ units

Direction $= \tan^{-1}\left(\frac{5}{2}\right)$ with the -ve x-axis

c) $A+B$

Magnitude $= \sqrt{9+9}$

$= 3\sqrt{2}$ units

Direction $= 45°$ with +ve x-axis

Q3 -

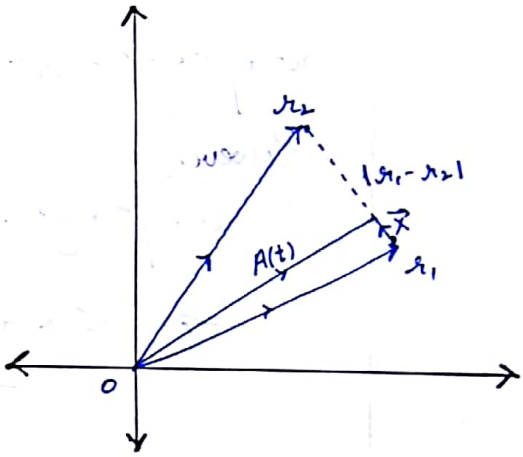
$\vec{A} = \vec{r_1} + \vec{x}$

$\frac{d\vec{A}}{dt} = \frac{d r_1}{dt} + \frac{dx}{dt}$

$= \frac{d(x\hat{x})}{dt}$

$= \hat{x} \cdot v$

$= v \cdot \frac{\vec{r_2} - \vec{r_1}}{|\vec{r_2} - \vec{r_1}|}$



$\int_{r_1}^{A} dA(t) = \int_{t_1}^{t} v \cdot \frac{\vec{r_2} - \vec{r_1}}{|\vec{r_2} - \vec{r_1}|} dt$

$\vec{A}(t) - \vec{r_1} = v \cdot \frac{\vec{r_2} - \vec{r_1}}{|\vec{r_2} - \vec{r_1}|} (t - t_1)$

At $t = t_2$, $A(t) = r_2$, $v = \frac{|\vec{r_2} - \vec{r_1}|}{t_2 - t_1}$

$A(t) = \vec{r_1} + \frac{\vec{r_2} - \vec{r_1}}{\cancel{|\vec{r_2} - \vec{r_1}|}} \cdot \frac{\cancel{|\vec{r_2} - \vec{r_1}|}}{t_2 - t_1} \cdot t - t_1$

$\therefore A(t) = \vec{r_1} + \frac{\vec{r_2} - \vec{r_1}}{T} \cdot (t - t_1)$

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b) $P = 300\,W$

$t = 8$ Hrs

Cost/unit = ₹2

No. of units consumed by 1 lamp $= 2400 \times 10^{-3}$
$= 2.4$

∴ Total no. of units consumed $= 2.4 \times 500$
$= \underline{\underline{1200}}$

Hence, total cost = ₹2400/-

∴ Per year $= 2400 \times 365$
$= \underline{\underline{₹\ 8,76,000/-}}$

Q3-

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TUTORIAL # 4

Q1 -

$$\text{Avg velocity} = \frac{\text{Total displacement}}{\text{Total time taken}}$$

$$= \frac{s_f - s_i}{t_f - t_i}$$

$$= \frac{20}{20} = \underline{\underline{0.1 \text{ m/s}}}$$

* we can find region of max. velocity as the region where slope is max as $v = \frac{ds}{dt}$ = slope

Hence, v_{max} is from $t = 10$ to $t = 16$

$$v_{max} = \frac{s_{16} - s_{10}}{6}$$

$$= \frac{1.8 - 0.4}{6}$$

$$= \frac{1.4}{6} = \underline{\underline{0.233 \text{ m/s}}}$$

Q2 -

$\dot{r} = 4 \text{ m/s}$

$\dot{\theta} = 2 \text{ rad/s}$

$r = 3 \text{ m}$

$\vec{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta}$

$= 4\hat{r} + 6\hat{\theta}$

$|\vec{v}| = \sqrt{16 + 36} = \underline{\sqrt{52} \text{ units}}$

$\vec{a} = [\ddot{r} - r\dot{\theta}^2]\hat{r} + (2\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\theta}$

$= -12\hat{r} + 16\hat{\theta}$

$|\vec{a}| = \sqrt{144 + 256}$

$= \underline{\underline{20 \text{ units}}}$.

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Q3- $F = (4t^2\hat{i} - 3t\hat{j})\,N$

$m = 5kg$

$F = ma$

$a = \frac{4t^2}{m}\hat{i} - \frac{3t}{m}\hat{j}$

$a = \frac{dv}{dt}$

$\therefore \int dv = \int a\,dt$

(i) $\Rightarrow \vec{v} = \frac{4t^3}{3m}\hat{i} - \frac{3t^2}{2m}\hat{j}$

$\vec{v} = \frac{4t^3}{15}\hat{i} - \frac{3t^2}{10}\hat{j}$

(ii) $v = \frac{dr}{dt}$

$\int dr = \int v\,dt$

(ii) $\therefore \vec{r} = \frac{4t^4}{60}\hat{i} - \frac{3t^3}{30}\hat{j}$

$\Rightarrow \vec{r} = \frac{t^4}{15}\hat{i} - \frac{t^3}{10}\hat{j}$

(iii) $\vec{r} \times \vec{v} = \begin{vmatrix} t^4/15 & -t^3/10 \\ 4t^3/15 & -3t^2/10 \end{vmatrix}$

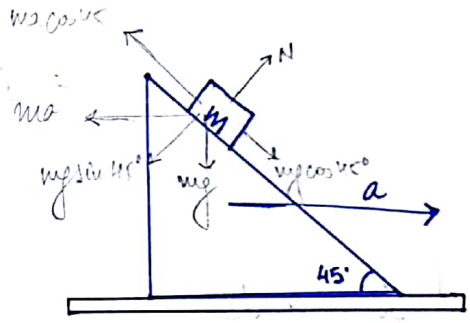
$= \frac{-3t^6}{150} + \frac{4t^6}{150}$

$= \frac{t^6}{150}$

Q4- $F = mg\cos 45 - ma\sin 45$

$\therefore m a_{block} = \frac{mg}{\sqrt{2}} - \frac{ma}{\sqrt{2}}$

$\Rightarrow a_{block} = \frac{1}{\sqrt{2}}(g - a)$



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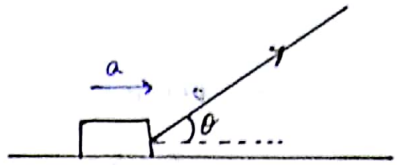
Tutorial #5

Q1- $F\cos\theta = ma$ —(I)

$[\because \Sigma F = ma]$

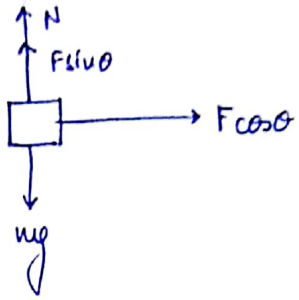


$F\sin\theta + N = mg$

$F\sin\theta = mg - N$ —(II)

(II) ÷ (I)

$\tan\theta = \dfrac{mg - N}{ma}$ —(III)



$F = \dfrac{ma}{\cos\theta}$

$= \dfrac{mg - N}{\sin\theta}$ $[\because$ From (III)$]$

$N = mg - F\sin\theta$

$N = mg - ma\tan\theta$

Q2- Let height at $T_1 = h$

$h = ut + \frac{1}{2}gt^2$

~~$h = 4.4\,T_2^2$~~

$h = -ut_2 + \frac{1}{2}gT_2$ —(II)

Now,

$ut = \dfrac{h}{T_1}$ —(I)

Sub (I) in (II)

$h = \dfrac{-hT_2}{T_1} + \frac{1}{2}gT_2^2$

$h\left(1 + \dfrac{T_2}{T_1}\right) = \frac{1}{2}gT_2^2$

$h = \dfrac{1}{2}\dfrac{gT_1T_2^2}{(T_1 + T_2)}$

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Tut #5

Q3 - From FBD I,

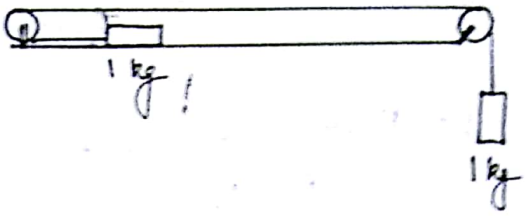
$10 - T = a$

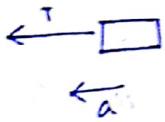
From FBD II,

$T = a$

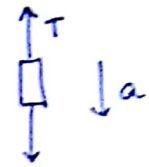
$\therefore 2T = 10$

$\Rightarrow$ $\underline{\underline{T = 5N}}$





II



I

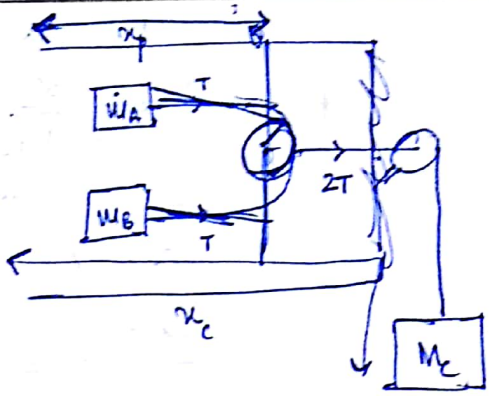
Q4 - $M_c g - 2T = M_c a_c$

For M_a

$M_a a_a = T$

For M_B

$M_B a_B = T$



Now,

$a_a = \frac{T}{M_a}$; $a_b = \frac{T}{M_b}$; $a_c = g - \frac{2T}{M_c}$

Constraint: $x_c - x_p = 0 \Rightarrow \ddot{x}_c - \ddot{x}_p = 0$

$\therefore a_c - a_p = 0$

$\therefore a_c = a_p$

$(x_p - x_A) + (x_p - x_B) + \pi R = \text{Constant}$

$\ddot{x}_p - \ddot{x}_a + \ddot{x}_p - \ddot{x}_B = 0$

$a_p - a_a + a_p - a_B = 0$

$2a_c = a_a + a_B$

$2g - \frac{4T}{M_c} = \frac{T}{M_a} + \frac{T}{M_B}$

$2g = T\left(\frac{4M_AM_B + M_BM_c + M_AM_c}{M_AM_BM_c}\right)$

$T = \frac{2g\, M_AM_BM_c}{4M_AM_B + M_BM_c + M_AM_c}$

$\Rightarrow a_a = \frac{2g\, M_BM_c}{4M_AM_B + M_BM_c + M_AM_c}$

$a_b = \frac{2g\, M_aM_c}{4M_AM_B + M_BM_c + M_AM_c}$

$a_c = g - \frac{4M_AM_B}{4M_AM_B + M_BM_c + M_AM_c}$

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Tutorial #6

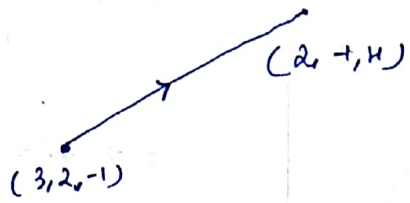
A1:

Initial position : $(3, 2, -1)$

$\vec{r_1} = 3\hat{i} + 2\hat{j} - \hat{k}$

Final position : $(2, -1, 4)$

$\vec{r_2} = 2\hat{i} - \hat{j} + 4\hat{k}$



Displacement, $\vec{s} = \vec{r_2} - \vec{r_1}$

$= -\hat{i} - 3\hat{j} + 5\hat{k}$

Given, $\vec{F} = 4\hat{i} - 3\hat{j} + 2\hat{k}$

we know,

$W = \vec{F} \cdot \vec{s}$

$= (4\hat{i} - 3\hat{j} + 2\hat{k}) \cdot (-\hat{i} - 3\hat{j} + 5\hat{k})$

$= -4 + 9 + 10$

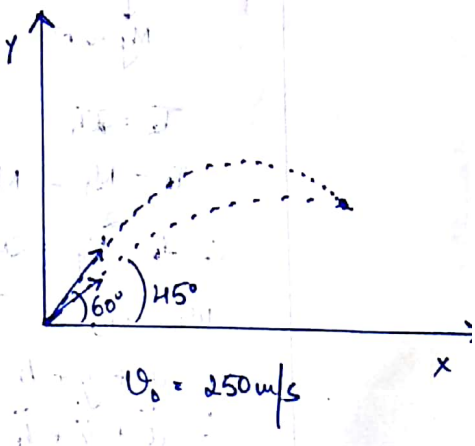
$= \underline{\underline{15 \text{ J}}}$

A2:

$x_A = x_B$

$\therefore \frac{250}{\sqrt{2}} t_2 = \frac{250}{\sqrt{2}} t_1$

$t_1 = \sqrt{2}\, t_2$ — (1)



$y_A = y_B$

$\therefore \frac{250}{\sqrt{2}} t_2 - \frac{1}{2} g t_2^2 = 250 \cdot \frac{\sqrt{3}}{2} t_1 - \frac{1}{2} g t_1^2$ — (4)

$\therefore$ Sub (1) in (4)

$\frac{250}{\sqrt{2}} t_2 - \frac{1}{2} g t_2^2 = 125\sqrt{3} \cdot \sqrt{2}\, t_2 - \frac{1}{2} g \cancel{2} t_2^2$

$-\frac{250}{\sqrt{2}} t_2 + 125\sqrt{6}\, t_2 = \frac{1}{2} g t_2^2$

$5 t_2^2 = 125\sqrt{6}\, t_2 - \frac{250}{\sqrt{2}} t_2$

As $t_2 \neq 0$

$5 t_2 = 125\sqrt{6} - \frac{250}{\sqrt{2}}$

$t_2 = 25\sqrt{6} - \frac{50}{\sqrt{2}}$

$\therefore t_2 = 25\sqrt{6} - 25\sqrt{2}$

$= 25(\sqrt{6} - \sqrt{2})$

$= 25(2.45 - 1.41)$

$= 25 \times 1.04$

$= 26$

$\therefore t_1 = 26\sqrt{2}$

$\Delta t = 26(\sqrt{2} - 1)$

$= 26 \cdot (0.414)$

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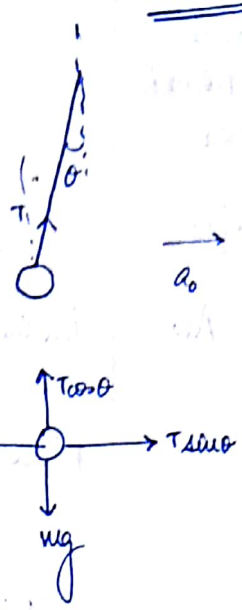
A3:

We see,

$T\cos\theta = mg$ —(I)

$T\sin\theta = ma_0$ —(II)

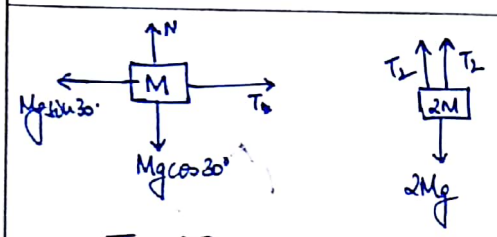


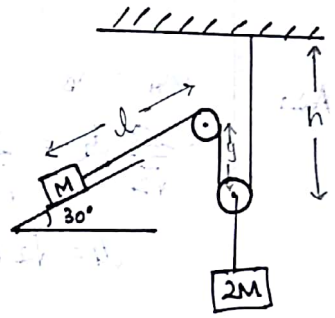
(II) ÷ (I)

$\tan\theta = \frac{a_0}{g}$

$\Rightarrow \theta = \tan^{-1}\left(\frac{a_0}{g}\right)$

A4:





$T_2 = 2T_1$

$T_1 - \frac{Mg}{2} = Ma_M$; $2Mg - 2T_1 = 2Ma_{2M}$

$\therefore 2T_1 - Mg = 2Ma_M$; $2Mg - 2T_1 = 2Ma_{2M}$

—(I) —(II)

We see,

$l + y + h = c_1$

$\therefore \ddot{l} + \ddot{y} + \ddot{h} = 0$

$\Rightarrow a_M + a_{2M} = -\ddot{y}$ —(III)

And,

$h - y = c_2$

$\therefore \ddot{h} = \ddot{y}$

$\Rightarrow a_{2M} = \ddot{y}$

Sub in (III)

$|a_M| = 2|a_{2M}|$

Plugging into (I) and (II)

$2T_1 - Mg = 4Ma_{2M}$ —(IV)

$2Mg - 2T = 2Ma_{2M}$ —(V)

(IV) + 2(V)

$3Mg = 6Ma_{2M}$

$\cancel{Mg = \frac{5}{2}Ma_M}$

$\cancel{\therefore a_M = \frac{2}{5}g}$

$Mg = 6Ma_{2M}$

$\therefore a_{2M} = \frac{g}{6}$

$\therefore a_M = g/3$

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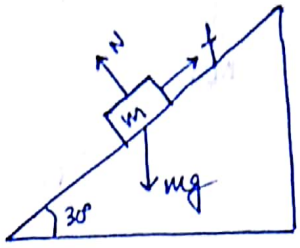
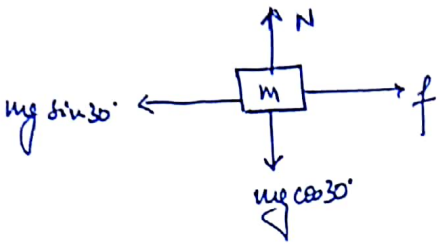
Tutorial # 7

A1:

$m = 3kg$

$s = 2m$





We see,

$N = mg\cos 30°$

$= 3 \times 10 \times \frac{\sqrt{3}}{2}$

$= 15\sqrt{3}\,N$

$f = mg\sin 30° = ma$

$f = \mu_k N$

$\mu_k = \frac{f}{N}$

$t = 1.5s$

$s = 2m$

$s = ut + \frac{1}{2}at^2$

$2 = \frac{1}{2} a \times (2.25)$

i) $a = \frac{4}{2.25}$

$\therefore a = 1.8\,m/s^2$

ii) $\therefore f = 3(1.8) + \frac{30}{2}$ $-3(1.8)$

$\therefore f = 15 - 5.4$

$= 9.6N$

iii) $\mu_k = \frac{9.6}{15\sqrt{3}}$

$= \frac{9.6}{25}$

$\Rightarrow \mu_k \simeq 0.37$

d) $v^2 = 2as$

$= 2 \times 1.8 \times 2$

$= 1.8 \times 4$

$\Rightarrow v \simeq 7.2\,m/s$

$\therefore v = 2.68\,m/s$

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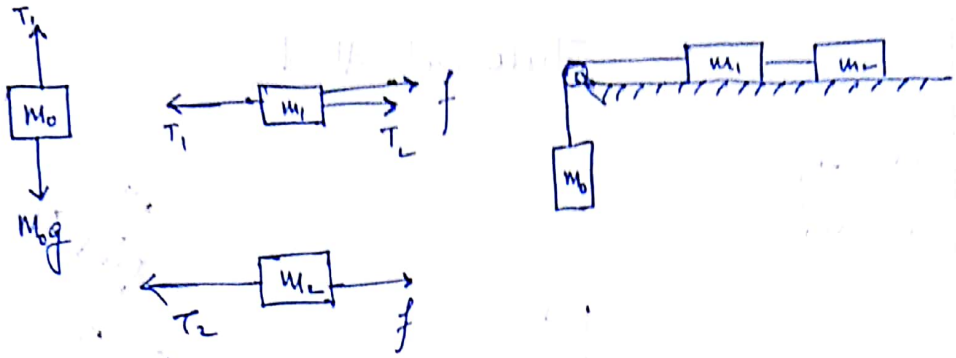
A2.



$\Sigma F = ma$

$m_0 g - T_1 = m_0 a$

$T_1 - T_2 - k m_1 g = m_1 a$

$T_2 - k m_2 g = m_2 a$

$T_1 = m_0(g-a)$

$T_2 = m_2(kg+a)$

$T_1 = m_1 a + k m_1 g + m_2(a+kg)$

$= m_1(a+kg) + m_2(a+kg)$

$= (m_1+m_2)(a+kg)$

$\therefore (m_1+m_2)(a+kg) = m_0(g-a)$

$2\cancel{m}(a+kg) = \cancel{m}(g-a)$

$\therefore 2a + 2kg = g - a$

$3a = g(1-2k)$

$\Longrightarrow a = \frac{g}{3}(1-2k)$

$T = m_2(kg+a)$

$= m\left(kg + \frac{g}{3} - \frac{2kg}{3}\right)$

$\therefore T = \frac{mg}{3}(k+1)$

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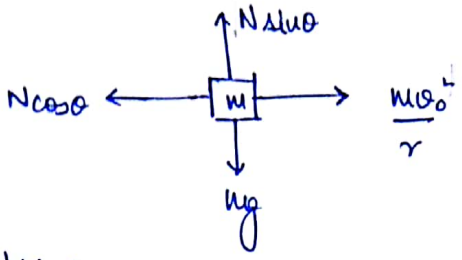
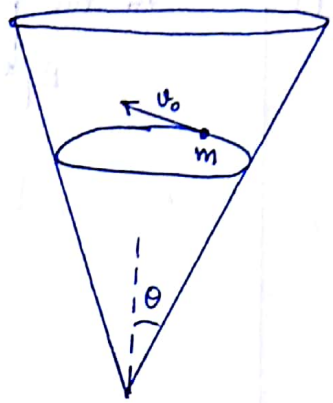
PHY - 101

Tutorial #8

21/9/17

A1:





$N\sin\theta = mg$ — (I)

$N\cos\theta = \frac{mv_0^2}{r}$ — (II)

(I) ÷ (II)

$\tan\theta = \frac{gr}{v_0^2}$

$\Rightarrow r = \frac{v_0^2 \tan\theta}{g}$

A2:

$r_A + r_B = l$

$\therefore \ddot{r}_a + \ddot{r}_B = 0$

$\Rightarrow \ddot{r}_a = -\ddot{r}_B$

$M_a\omega^2 r_a - T = M_a\ddot{r}_a$ — (I)

And,

$T - M_B\omega^2 r_B = M_B\ddot{r}_B$

$\therefore T - M_B\omega^2 r_B = -M_B\ddot{r}_a$ — (II)

(I) + (II)

$(M_A r_a - M_B r_B)\omega^2 = (M_A - M_B)\ddot{r}_a$

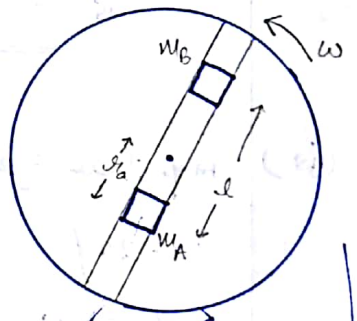
$\therefore [M_A r_a - M_B(l - r_A)]\omega^2 = (M_A - M_B)\ddot{r}_a$

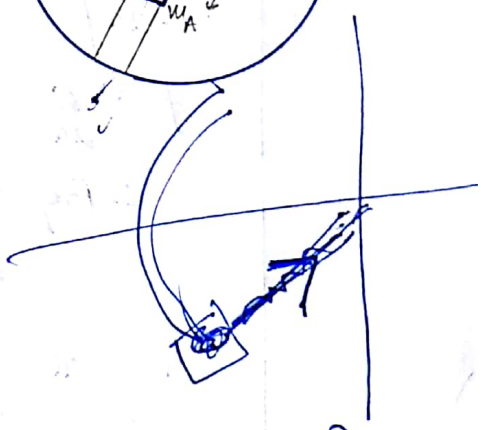
$\Rightarrow \ddot{r}_a = \frac{[M_A r_A - M_B(l - r_A)]\omega^2}{(M_A - M_B)}$

$-T = M_a(\ddot{r}_a - r_a\omega^2)$

$-T = M_a(\ddot{r}_B - r_B\omega^2)$

Solving, we get the final ans.





$m\ddot{r}_B + \left(-\frac{mv^2}{r_B}\right) = -T$

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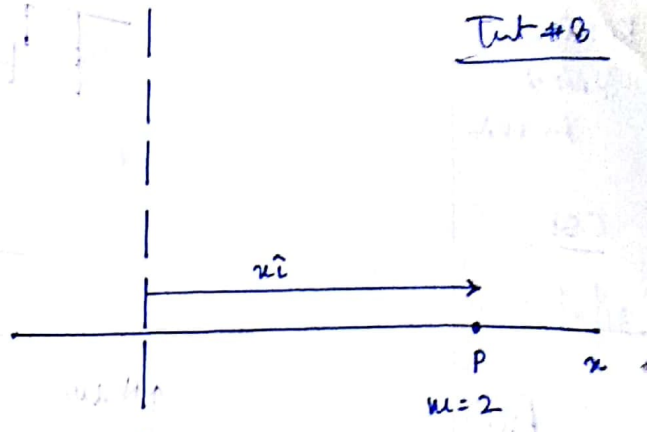
Tut #8

A3: $x = 20$

$F = -8x\hat{i}$

$F = -kx$

$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{8}{2}} = 2$



We know,

(i) As the particle is in SHM

Eq^n →

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

$$\therefore \frac{d^2x}{dt^2} + 4 = 0$$

(ii) Let $x = A\cos\omega t + B\sin\omega t$

We know, At $t = 0$, $x = 20$

$\therefore A = 20$, B

$v = -\omega A\sin\omega t + B\omega\cos\omega t$

At $t = 0$, $v = 0$ $\therefore$ $0 = -2(20)(0) + B(2)(1)$

$\therefore B = 0$ $\Rightarrow x = 20\cos\omega t$

(iii) $v = \dot{x} = -20\omega\sin\omega t$

$\vec{v} = -40\sin 2t\,\hat{i}$

Speed = $40\sin 2t$

(iv) Amplitude = 20

$\omega = 2\pi f$

$f = \frac{\omega}{2\pi}$

$\therefore$ Freq $= \frac{1}{\pi}$

$T = \pi$

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Tut # 8

Ans:

$r = 5 \times 10^{-4}$ m

$m = 10^{-3}$ kg

$\eta = 0.36$ Ns/m^2

Terminal velocity is the const. velocity a body acquires when it falls through a viscous liquid as the gravitational force becomes equal to the viscous force and hence, net force and acceleration = 0.

$\therefore$ Viscous force = Gravitational force

$6\pi\eta r v = mg$ [$\because$ From formula, $F_v + F_b = mg$]

$$v_T = \frac{mg}{6\pi\eta r}$$

$$= \frac{\cancel{10^{-3}} \times 9.8}{6 \times 3.14 \times 0.36 \times \cancel{10^{-3}} \times 0.5}$$

$$= \frac{9.8}{0.36 \times 3\pi}$$

$$\Longrightarrow v_T = \underline{\underline{2.88 \text{ m/s}}}.$$

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